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Distribution and potential health risks of perfluoroalkyl substances (PFASs) in water, sediment, and fish in Dongjiang River Basin, southern China

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Abstract

Per- and polyfluoroalkyl substances (PFASs) have attracted worldwide attention due to their high stability, refractory degradation, and bioaccumulation. In this study, 17 PFASs in water, sediment, and fish in the Dongjiang River Basin in southern China were investigated using high-performance liquid chromatography-mass spectrometry. Total PFASs concentrations ranged from 20.83 to 372.8 ng/L in water, from 1.050 to 3.050 ng/g in sediments, and from 12.28 to 117.4 ng/g in fish. Among six species of fish, Oreochromis mossambicus (mean: 68.55 ng/g) had the highest concentration of PFASs, while Tilapia zillii (36.90 ng/g) had the lowest concentration. Perfluorooctanoic acid (PFOA) predominates in water and sediments, while perfluorooctanesulfonic acid (PFOS) predominates in fish. Long-chain perfluorocarboxylates (PFCAs) and perfluorosulfonates (PFSAs) showed higher bioaccumulation, and the field-sourced sediment-water partition coefficients (K_d) and bioaccumulation factors (BAFs) of PFASs increased with the length of perfluorocarbon chains. PFASs concentration in the lower reaches (urban area) of the Dongjiang River is higher than that in the upper and middle reaches (rural area). The calculated hazard ratio (HR) of PFOS and PFOA levels in fish in the Dongjiang River Basin were far less than 1, hence the potential risk to human health was limited.

1. Introduction

Per- and polyfluoroalkyl substances (PFASs) are synthetic organic compounds in which the hydrogen atoms on the carbon chain have been partially or completely replaced by fluorine atoms. Carbon-fluorine bonds have extremely high bond energy (Buck et al., 2011), so PFASs have high thermal stability, refractory degradation, and high durability. In addition, owing to their high surface activity, hydrophobicity, and oleophobicity, PFASs are widely used in human daily life, such as surfactants, fireproof foam, and food packaging (Ahrens and Bundschuh, 2014; Giesy and Kannan, 2001; Route et al., 2014; Trier et al., 2011). Generally, PFASs can be classified into various categories, mainly including perfluorocarboxylic acids (PFCAs), perfluorosulfonic acids (PFSAs), perfluorotelomer alcohols (FTOHs), etc. (Wang et al., 2011). Perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) as the representative substances of PFCAs and PFSAs, were listed in the Stockholm Convention and Regulations on Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH), and perfluorohexylsulfonic acids (PFHxS) are proposed to be included in the annex list (POPRC, 2018; UNEP, 2019). With the phasing out of traditional PFASs, PFAS production in China has gradually shifted to short-chain PFASs and structurally similar fluorinated alternatives, such as perfluorohexanecarboxylic acid (PFHxA), perfluorobutanesulfonic acid (PFBS), and fluoroethersulfonic acid.

Numerous studies have shown that PFASs are frequently detected in a variety of environmental media, such as water, air, soil, dust, and sediment (Ahmed et al., 2020; Pan et al., 2014a; Wang et al., 2015; Yao et al., 2018; Zhao et al., 2016b). Ao et al. (2019) reported that the concentration range of six typical PFASs in drinking water was 0.310–4.14 ng/L. Meng et al. (2019) found PFCAs and PFOS were the predominated compounds in the sediments from the Guanting Reservoir in Zhangjiakou. Moreover, PFASs can also bioaccumulate in organisms through the food chain, causing potential harm to humans (Lee et al.,

2020a; Li et al., 2013). PFASs have been found to cause serious health effects in humans, including multiorgan toxicity, neurotoxicity, reduced infertility, developmental toxicity, immune toxicity, and even cancer (Zeng et al., 2019). Dietary intake is one of the most important ways to human exposure to PFASs, especially from fish (Fair et al., 2019; Haug et al., 2010). Hung et al. (2020) reported PFASs concentrations ranging from 0.99 to 3.04 ng/g (wet weight) in edible fish.

The Dongjiang River is one of the main tributaries of the Pearl River and one of the most important water sources in the Pearl River Delta. The Dongjiang Valley is the most developed area in the Greater Bay Area (Guangzhou, Shenzhen, Dongguan, Hong Kong). Meanwhile, the fish breeding industry in the area is developed, and fish consumption is higher than other regions. Kannan et al. (1997) reported that edible fish is one of the main ways to expose human organic pollutants in the Pearl River Delta region. Due to the high level of urbanization in the middle and lower sections of the Dongjiang River, human exposure to PFASs is higher there. The concentration of PFASs is directly correlated with the degree of industrialisation and urbanization (Pan et al., 2014a). However, there are few studies on the spatial distribution of PFASs in the whole Dongjiang River and the changes of PFASs transfer between different media in the river ecology of Dongjiang River (Liu et al., 2015; Pan et al., 2014a; Zhang et al., 2013). Liu et al. (2015) investigated the concentration of PFASs in the surface water on the Pearl River, it was found that the PFAS content in the samples of developed cities in the Pearl River Basin was higher. However, the transfer, fate, distribution and potential harm of PFASs to residents along the Dongjiang River ecosystem are not clear.

The purpose of this study was (1) to investigate the concentration and distribution levels of PFASs in water, sediments, and fish samples from the Dongjiang River; (2) to assess their potential risk to coastal residents; (3) to explore the migration, transformation, and bioaccumulation of PFASs in the Dongjiang River Basin; (4) to provide data support for the formulation of PFASs pollution control strategies for drinking water sources such as Xinfengjiang Reservoir and Fengshuba Reservoir.

2. Materials And Methods

2.1. Chemicals

A total of 17 PFASs standard solutions (perfluorobutanoic acid (PFBA), per fluoropentanoic acid (PFPeA), PFHxA, perfluoroheptanoic acid (PFHpA), PFOA, perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUdA), perfluorododecanoic acid (PFDoA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), PFBS, PFHxS, perfluoroheptane sulfonic acid (PFHpS), PFOS, perflurodecanesulphonate (PFDS), perfluorohexylphosphonic acid (PFHxPA)) were purchased from Wellington Laboratory (Guelph, Ontario, Canada). Recovery indicators for PFASs analysis using two isotopically labeled standards (13C2-PFOA and 13C8-PFOS) and a mixed internal standard (MPFAC-max) (including MPFBA (13C4), MPFHxA (13C2), MPFOA (13C4), MPFNA (13C5), MPFDA (13C2), MPFUndA (13C2), MPFDoA (13C2), MPFHxS (18O2), MPFOS (13C4)) for quantification, provided by Wellington Laboratories (Guelph, Ontario, Canada). Ultrapure water was used throughout the study, produced by a

Milli-Q purification system (Millipore, Germany). The details of other chemical reagents in this experiment are shown in Table S1.

2.2. Sample collection and processing

The Dongjiang River is one of the main streams in the Pearl River system, with a total length of 562 kilometers. It originates in Xunwu County, Jiangxi Province, China, and flows through Guangdong Province, and Hong Kong. The Dongjiang Basin has two large reservoirs (Xinfengjiang Reservoir and Fengshuba Reservoir), and a Dongjiang-Shenzhen Water Supply Project, which supplies water to Hong Kong. It is the main source of drinking water for more than 40 million people in major cities such as Hong Kong, Guangzhou, and Shenzhen. In recent years, the development of the Guangdong-Hong Kong-Macao Greater Bay Area has been deepened and the level of urbanization has been continuously improved.

From December 6th to 10th, 2021, sample collection activities were carried out in the Dongjiang River Basin in southern China, and 34 sampling sites were set up (Fig. 1, Table S2). Along the river, 34 water samples, 33 sediment samples, and 81 fish samples were collected. Six types of fish samples were collected, namely Redbelly tilapia, Xenocypris microlepis, Opsariichthys bidens, Hemiculter leucisculus, Cirrhinus molitorella, and Oreochromis mossambicus. A 500 mL water sample was collected and stored in a Polypropylene (PP) bottle. The bottle was rinsed three times with methanol and ultrapure water before use and then stored in a cool place. Sediments were collected with grab dredgers and preserved in PP-sealed bags. At each sampling location, fish samples were collected, measured for body weight and length, sealed in PP bags, and kept refrigerated at -18°C. Immediately after the sampling activities, samples were sent to the laboratory. The fish samples were thawed, viscera were removed, and the weight was recorded, then wrapped in tinfoil and freeze-dried using a freeze dryer, and the weight was recorded again (Table S2). Finally, the fish samples were ground with a grinder, sealed in PP bags, and stored in a -18° refrigerator. The sediment was lyophilized using a lyophilizer, then uniformly ground, sealed in a polyethylene bag, and stored at room temperature. All equipments were rinsed with methanol before use.

2.3. Sample extraction and analysis

A 3 ng recovery indicator was applied after 100 mL of water sample had been filtered through a 0.22 μm nylon organic phase filter into a PP tube. Firstly, the Oasis WAX (60mg, 3cc; Waters, Milford, MA, USA) cartridge was preconditioned with 4 mL of 0.1% NH_4OH in methanol, 4 mL of methanol, and 4 mL of Milli-Q water. Cartridges were rinsed with 4 mL of 25 mM ammonium acetate (pH = 4), then drained for 30 min, followed by eluting and collection with 4 mL of methanol and 4 mL of 0.1% NH_4OH in methanol. The eluent was blown to near dry with high-purity nitrogen, then methanol was added and rinsed with 0.22 μm nylon organic phase filter into the LC vials. Finally, 2.5 ng internal standard sample was added to 0.5 mL for LC-MS-MS analysis.

A 0.5 g of a dry weight sediment sample was weighed into a 50 mL PP centrifuge tube and added 5 ng of internal standard, add 2 mL of Milli-O water, and vortex for 5 min. Add 2 mL of 0.25 M sodium carbonate (Na_2CO_3) solution and 1 mL of 0.5 M tetrabutylammonium bisulfate hydrogen sulfate (TBAHS) to each

centrifuge tube and vortex for 5 min. Then the tubes were placed in a preheated ultrasonic bath (60°C) and sonicated for 30 min. After sonication, 4 mL of Methyl tert-butyl ether (MTBE) was added and shaken for 20 min. After centrifugation at 3500 rpm for 30 min, the supernatant was transferred to a 15 mL PP centrifuge tube. A 5 mL of MTBE was added to the remaining aqueous solution again, and shaken and centrifuged the supernatant under the same conditions before transferring the supernatant to a 15 mL PP centrifuge tube. Repeated the above steps twice and combined the supernatant in a 15 mL PP tube. The supernatant was evaporated to dryness under a stream of nitrogen and redissolved with 0.5 mL of methanol. The as-received solution was filtered through a 0.22 μm nylon filter into a PP injection vial and analyzed using an LC-MS-MS instrument.

A 0.1 g of fish sample, 3 ng of recovery indicator, and 2 mL of sodium carbonate solution (0.25 M) were added to a 15 mL PP tube accordingly. Then 4 mL of MTBE solution was added into the tube, and the mixture was shaken for 30 min and centrifuged at 4000 rpm for 5 min. After that, the supernatant was transferred to another 15 mL centrifuge tube. Then 5 mL of MTBE solution was added to the sample for repeated extraction. The above steps were repeated three times to collect a total of 12 mL extracts. After blowing to 0.5 mL with moderate nitrogen flow, methanol solvent was added to control the volume of the solution to 1.0 mL. Finally, the mixture was purified by the Oasis WAX (60 mg/3 mL) cartridge after adding 5.0 mL of ultrapure water. The purification procedure was the same as the water sample, followed by the same steps as the water sample to prepare for LC-MS-MS analysis.

2.4. Instrumental analysis

PFASs in all the samples were analyzed using an LC-MS/MS system (AB SCIEX TRIPLE QUAD 5500) with a negative electrospray ionization (ESI) source and multiple reaction monitoring (MRM) mode. The separation of the individual PFAS was performed on a 100 mm × 2.1 mm Thermo scientific C18 column with a 5 μm particle size. 2 mM ammonium acetate (A) and HPLC grade methanol (B) were used as gradient mobile phases at a flow rate of 0.3 ml/min. Detailed procedures of instrumental analysis of PFASs are presented in Supporting Information, Table S11.

2.5. Quality control and quality assurance

Program blanks and solvent blanks were set for each batch of 10 samples to check for background signals. The mean of the overall blank was subtracted from each sample data to correct for background levels. All target analytes were quantified using an isotope-labeled internal standard method. An 8-point linear calibration curve was created in the range of 0.5 to 100 ng/mL, and the calibration curves of all PFASs were close to linear with regression coefficients (r2) above 0.999. After each 10 samples, test a mixture of 10 ng/mL PFASs standard solution and methanol to confirm the calibration of the instrument. Matrix-spiked and blank spiked control experiments were conducted on the samples to test the reliability of the methods. Detailed recovery information of 17 PFASs was shown in Table S3.

2.6. Data analysis

Data processing was performed using Social Sciences (SPSS) version 26.0 (IBM, Chicago, IL, USA) software. Before processing the data, the kolmogorov-smirnov test was used to test the data for non-normal distribution. The correlations between concentrations of PFASs in the samples were estimated by calculating the Spearman rank correlation coefficient. $P < 0.05$ was considered statistically significant. Concentrations less than LOQ were calculated using LOQ/2.

3. Results And Discussions

3.1. PFASs in water samples

The species composition and concentrations of 17 PFASs in the Dongjiang River Basin during the dry season are shown in Fig. 2. The total PFASs concentrations in water samples from the Dongjiang River Basin ranged from 20.73 to 372.8 ng/L, with a median concentration of 50.72 ng/L (Fig. 2, Table S4). Compared to PFASs levels in other rivers, the Dongjiang River Basin has middle-level PFAS concentrations (Chen et al., 2021a; Meng et al., 2019; So et al., 2007; Zhao et al., 2020). Chen et al. (2021a) investigated the PFASs content in Xijiang and Beijiang during the dry and wet seasons, and found that the total PFASs concentration ranged from 0.775 to 441.0 ng/L in the dry season and from 2.66 to 1060 ng/L in the wet season. Li et al. (2022) investigated PFASs levels ranging from 11.8 to 281 ng/L in several tributaries of the Pearl River estuary in 2021. Among the 17 kinds of detected PFASs in this study, detection frequencies of PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFBS, PFHxS, PFOS, and PFHxPA in samples were all above 85%. The detection rate of long-chain PFASs (carbon chain length 8, 14.71–47.06%) was lower than short-chain PFASs (Table S4). In general, PFCAs concentration is higher than that of PFSAs in all water samples (Fig. 2). PFOA predominates in water samples (mean = 20.9 ng/L), which contrasts with the findings of PFAS investigations conducted in previous research. High levels of PFOA and PFOS are found in the Haihe River in the north, with PFOS predominating (Li et al., 2011; Pan et al., 2011; Wang et al., 2012). This may be because PFOS usage was previously banned and replaced by PFBS, while PFOA was steadily reduced in recent years (Muir and Miaz, 2021). However, PFSAs concentration in the lower reaches of the Dongjiang River (21–34 sites) showed an increasing trend compared with that of the middle and upper reaches, especially PFBS. The possible reason is the increased use of PFSAs downstream due to its proximity to industrial areas, including the electroplating industry.

Figure 2 depicts the content and composition of PFAS in 34 sampling points from upstream to downstream. PFAS concentrations in the lower reaches of the Dongjiang River was much higher than that in the upper reaches, indicating PFASs concentration is closely related to the degree of industrialization and urbanization (Pan et al., 2014a). PFAS concentrations in Site 32 is the highest (372.8 ng/L), and the possible reason is that this site was heavily polluted by industrial activities. Site 4, 10, 11, 12 and 20 are all reservoirs that provide domestic water for the surrounding residents. The concentrations of PFASs in Xinfengjiang Reservoir (site 4) and Baipenzhu reservoir (site 20) were 58.88 ng/L and 49.91 ng/L, respectively. PFASs concentrations in Fengshuba Reservoir (sites 10, 11, 12) ranged from 33.64 to 42.17 ng/L. Compared with other reservoirs, the three reservoirs had moderate levels of PFASs concentration. For example, Meng et al. (2019) investigated 12 PFASs concentrations in Guanting Reservoir (24.7–142

ng/L). Li et al. (2020) investigated 18 PFASs in Danjiangkou Reservoir with concentrations ranging from 0.46 to 97.94 ng/L.

3.2. PFASs in sediment samples

The compositions and concentration of 17 PFASs in the sediments of the Dongjiang River Basin are shown in Fig. 3. PFASs concentrations in the sediments ranged from 1.05 to 3.50 ng/g, with a median value of 1.63 ng/g (Fig. 3, Table S5). PFOA and PFOS concentrations ranged from 0.21–1.30 ng/g and 0.13–0.50 ng/g, the medians were 0.45 ng/g and 0.28 ng/g, respectively. Similar to the water samples, the PFASs in the sediments showed an increasing trend with the flow direction of the Dongjiang River basin. The possible reason is that the middle and upper reaches of the rivers are located in the mountains and jungles, with fewer human activities and less industrial impact. PFASs in sediments are dominated by long-chain PFCAs (mean = 1.07 ng/g), whereas shorter PFASs (mean = 0.18 ng/g) are less abundant. The possible reason for this phenomenon is that short-chain PFASs are more easily dissolved in water and thus more difficult to be adsorbed by sediment (Ahrens et al., 2009).

Comparing the composition of PFASs in water and sediment samples (Fig. 2 and Fig. 3), it can be found that PFOS concentration in water and sediment is much lower than that of PFOA, which is similar to other region (Hung et al., 2020; Zhou et al., 2013). In 2010, Bao et al. (2010) reported that PFOS was the dominated PFASs in sediments of Zhuhai. However, PFOA is the main pollutant in the Pearl River reach part of this study. It is speculated that the reason may be the restricted use of PFOS in the past decade, and further studies are needed to verify this phenomenon.

3.3. PFASs in fish samples

As shown in Fig. 4 and Table S6, 17 PFASs were detected in all fish samples. In this study, Oreochromis mossambicus had the highest concentration of PFASs (68.55 ng/g) and Xenocypris microlepis had the lowest concentration (43.36 ng/g). The detection frequency of PFOA, PFPeA, PFNA, PFDA, PFUdA, PFDoA, PFTrDA, PFOS, PFDS is 100%, and the others are over 80%: PFHxA (96.9%), PFHpA (96.9%), PFHpS (96.9%), PFHxPA (96.9%), PFBS (93.8%), PFTeDA (90.6%), PFBA (84.4%), PFHxS (81.3%) (Table S6). PFOS accounted for the highest proportion (15.09%-49.1%) in all fish samples, followed by PFUdA, PFDA, and PFTrDA. Similar to our findings, Hung et al. (2020) found PFOS dominates in the liver samples of gray mullet (111.8 ng/g) and the blood samples of Japanese amberjack (26.35 ng/g) in Mashan, South Korea. And the concentrations of PFOS were highest in Baiyangdian fish in northern China, ranging from 2.67 to 17.4 ng/g (Shi et al., 2012).

Compared with water samples and sediments, fish samples had a more varied composition of PFASs, with a greater proportion of long-chain PFASs and relatively fewer short-chain PFASs (Fig. 4).This is similar to the findings of Meng et al. (2019) in Shijiazhuang, which indicates that long-chain PFASs are easier to accumulate in *vivo* than short-chain PFASs, owing to the bioaccumulation and biomagnification effects (Jeon et al., 2010; Loi et al., 2011).

3.4. Correlation analysis and distribution behavior of PFASs in the water-sediment-fish system

Spearman correlation analysis showed that the main PFASs in water samples were significantly correlated (r = 0.338–0.762, $p < 0.05$), indicating that PFASs in water samples may have the same source of pollution (Fig. S2). Similar to the Korean coastal environment (Lee et al., 2020b), PFCAs were more correlated than PFSAs in this study, it may be that there are more PFCAs in water samples. The correlation between PFOA and PFPeA was the highest (r = 0.76), this indicates that PFOA and PFPeA may have the same origin. Correlations among PFASs in sediments were low; similar to water samples, there were significant correlations among long-chain PFCAs in fish (Table S10).

The accumulation of pollutants in water is expressed by bioaccumulation factors (BAF), which is defined as the ratio of PFASs concentration in fish (ng/g) to PFASs concentration in water (ng/mL). The Log BAF values of fish PFASs are shown in Table S8. Long-chain PFASs have a high Log BAF value, which suggests that they have considerable bioaccumulation. The Log BAF value of PFUdA (mean:3.77) is the highest, followed by PFDA (mean:3.76), PFDoA (mean: 3.47), PFTrDA (mean:3.60). This is similar to some studies, such as Zhangjiakou Yongding River Basin Biology, Taihu Lake Biology, Hong Kong Biology, and Pearl River Biology(Li et al., 2022; Loi et al., 2011; Meng et al., 2019; Xu et al., 2014), it was found that the Log BAF value of long-chain PFASs was larger than that of short-chain PFASs, which also indicated that long-chain PFASs were more likely to accumulate and aggregate in *vivo*.

The partition coefficients of each PFASs for water and sediments are expressed in K_d (Table S7). The Log K_d of the PFASs of C4-C14 ranges from 0.71 to 2.57, which is close to that of other rivers. The PFASs of C3-C13 in the Pearl River range from 0.52 to 2.87 (Li et al., 2022), and the PFASs of C4-C11 in the Bohai Sea are 0.66–2.9 (Chen et al., 2020), and 0.67 to 2.18 for River Xi C5-C11 PFASs (Chen et al., 2018). Fig. S1 shows that the Log Kd is linearly related to the carbon chain length of PFASs ($r^2 = 0.68$). Similar to the findings of several earlier investigations (Ahrens et al., 2010; Higgins and Luthy, 2006), log Kd rises as carbon chain length grows. This suggests that the distribution of PFASs in water and sediments is greatly influenced by the length of the carbon chain.

The BAF and bio-sediment accumulation factor (BSAF) were calculated according to the content of PFASs in sediments and organisms to evaluate the bioaccumulation potential of PFASs in sediments and fish, respectively, and to understand the interaction of PFASs with the surrounding environment. Same to the Log Kd value, both Log BAF ($r^2 = 0.68$) and Log BSAF ($r^2 = 0.02$) showed a positive correlation with carbon chain length, consistent with previous studies (Pan et al., 2021; Wang et al., 2020), indicating that the elastic chain length of PFASs is an important element affecting the bioaccumulation of PFASs. In this study, the Log BAF of fish ranged from 2.03 to 3.08, which was much larger than 1, indicating that PFASs have great potential for transfer from water bodies to fish. In addition, the concentration of PFASs in different matrices was different. For example, compared with water samples, the content of long-chain PFASs in sediments was significantly increased; in organisms, long-chain PFASs dominated.

The bioaccumulation of PFASs may vary between different organisms. Li et al. (2022) found that the BAF of PFOA in oysters was an order of magnitude larger than that in pelagic fish, while the BAF of PFOS in pelagic fish was an order of magnitude larger than oysters. In this study, it was found that under the same carbon chain length, PFOS (Log BAF:3.61;Log Kd:1.87) was higher than PFOA (Log BAF:2.13 Log Kd:1.43), and PFHxS (Log BAF:2.88; Log Kd:1.87) was also higher than PFHxA (Log BAF:2.54; Log Kd:1.01), indicating that PFSAs (mean Log BAF:3.02) may have stronger bioaccumulation than PFCAs (mean Log BAF:2.67) in fish. However, the bioaccumulation potential of long-chain PFCAs with carbon lengths from 10 to 14 is close to or even higher than that of PFOS, possibly because the adsorption capacity of PFASs for lipids and proteins also increases with the increase of carbon chain length (Conder et al., 2008). Therefore, the potential bioaccumulative property of long-chain PFCAs needs to be paid attention to. Additionally, this research discovered that when the lengths of the carbon chains were equal, Log BAF was the highest and Log BSAF was the smallest. This shows that of the three modes of transport, the transfer of PFASs from water to fish is the primary pathway. These results contribute to a better understanding of the transport of PFASs and are essential for modeling the migration and transformation behavior of PFASs in aqueous environments.

3.5. Human exposure risk assessment

Studies in medicine and epidemiology have shown that PFAS exposure can have negative effects on human health. Fish in the Dongjiang River are more prone to accumulate PFASs, according to the previous detection results. However, fish may be one of the major causes of PFAS contamination in humans (Berger et al., 2009; Domingo and Nadal, 2017; Ericson et al., 2008), because it is the primary source of protein and fatty acids consumed by humans (Berger et al., 2009; He et al., 2004). The estimated daily intake (EDI) and the hazard ratio (HR) are commonly used indicators to evaluate the potential impact of eating fish PFASs on human health.

$$EDI = C_{PFASs\ (ng/g)} \times DI_{(g/day)} / BW_{(kg)}$$

The hazard ratio = EDI / RfD, where RfD is the Reference Dose.

In this study, the per capita daily intake (DI) of fish and meat in the Chinese population (19.5 g/day) was obtained from the 5th China Total Diet Studies (TDS) (Wang et al., 2019). The mean human body weight (BW) was calculated using 63 Kg, as same as the previous population survey data (Ericson et al., 2008; Wu et al., 2012). Based on previous studies, the Reference Dose (RfD) values for PFOS and PFOA were set at 150 and 1500 ng/kg/d, respectively (Authority, 2008). If the HR value is greater than 1, it indicates that there is a harmful risk. The EDI and HR values of typical PFASs in this experiment are shown in Table 1. The mean EDI value of PFOS in this study (5.67 ng/kg/day) was dominant among all PFASs, which was higher than that in some regions, such as Sweden (0.62 ng/kg/day), Norway (0.78 ng/kg/day) kg/day), Taihu Lake (1.61 ng/kg/day), and Hong Kong (2.4 ng/kg/day) (Berger et al., 2009; Chen et al., 2021b; Haug et al., 2010; Zhou et al., 2014). However, compared with the previous Pearl River region (31 ng/kg/day), it was much lower (Pan et al., 2014b). The HRs of PFOS and PFOA in fish in the Dongjiang River Basin were far less than 1, and the potential for harm to human health was limited.

Table 1
EDI and HR values of typical PFASs in different fish from the Dongjiang River.

Fish name	EDI (ng/kg/day)		Hazard ratio	
	PFOA	PFOS	PFOA	PFOS
Xenocypris microlepis	0.49	3.44	3.29×10^{-4}	0.016
Cirrhinus molitorella	1.33	2.15	8.86×10^{-4}	0.045
Opsariichthys bidens	1.50	2.88	1.00×10^{-4}	0.023
Tilapia zillii	1.58	4.62	1.05×10^{-4}	0.013
Hemiculter leucisculus	0.54	3.72	3.58×10^{-4}	0.032
Oreochromis mossambicus	0.79	6.20	5.24×10^{-4}	0.053

4. Conclusions

This study presents a general overview and detailed analysis of PFASs concentrations in water, sediment, and fish samples from the Dongjiang River Basin in southern China. In the Dongjiang River basin, PFAS concentrations in water and sediments generally rose with river flow, and the contamination level was higher downstream than upstream, particularly in the Shima and Freshwater River basins. Among the water samples, short-chain PFASs such as PFBA, PFPeA, PFHxA and PFHpA were predominated, and PFOA accounted for the highest proportion. PFASs with long chains, such as PFOA, PFOS, and PFNA, predominate in sediments. PFASs have bioaccumulation effects in fish samples, and long-chain PFASs and PFSAs are more bioaccumulative than short-chain PFASs and PFCAs. The Kd, BAF and BSAF values of PFASs are positively correlated with the carbon chain length. Both the carbon chain length and functional groups are important factors affecting the distribution and bioaccumulation of PFASs water ash, which provides data support for the transfer and transformation of PFASs in water environment. Human exposure risk assessments indicate that the concentration levels of PFOS and PFOA, prevalent PFAS contaminants in fish, are less harmful to people living in the Dongjiang River Basin.

Declarations

Data availability

All data generated or analyzed during this study are included in this published article.

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Author Contributions

All authors contributed to the study conception and design. The investigation, sampling, methodology and writing were completed by Shibo Li and Wang Xu. Sampling was done by Weimin Wang, Ping Sun, Liang Yu, Chunyang Yin, Xuxia Li and Gang Ren. The polishing, editing and revision were done by Lin Peng. The revisions and edits were done by Fei Wang.

Ethics declarations

Ethical approval

The use of fishes was approved by the Ethics Committee of Jinan University, and governmental regulations about the ethical use of fishes were followed.

Consent to participate

All authors participated in this work.

Consent for publication

All authors agree to publish.

Competing interests

The authors declare no competing interests.

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Figures

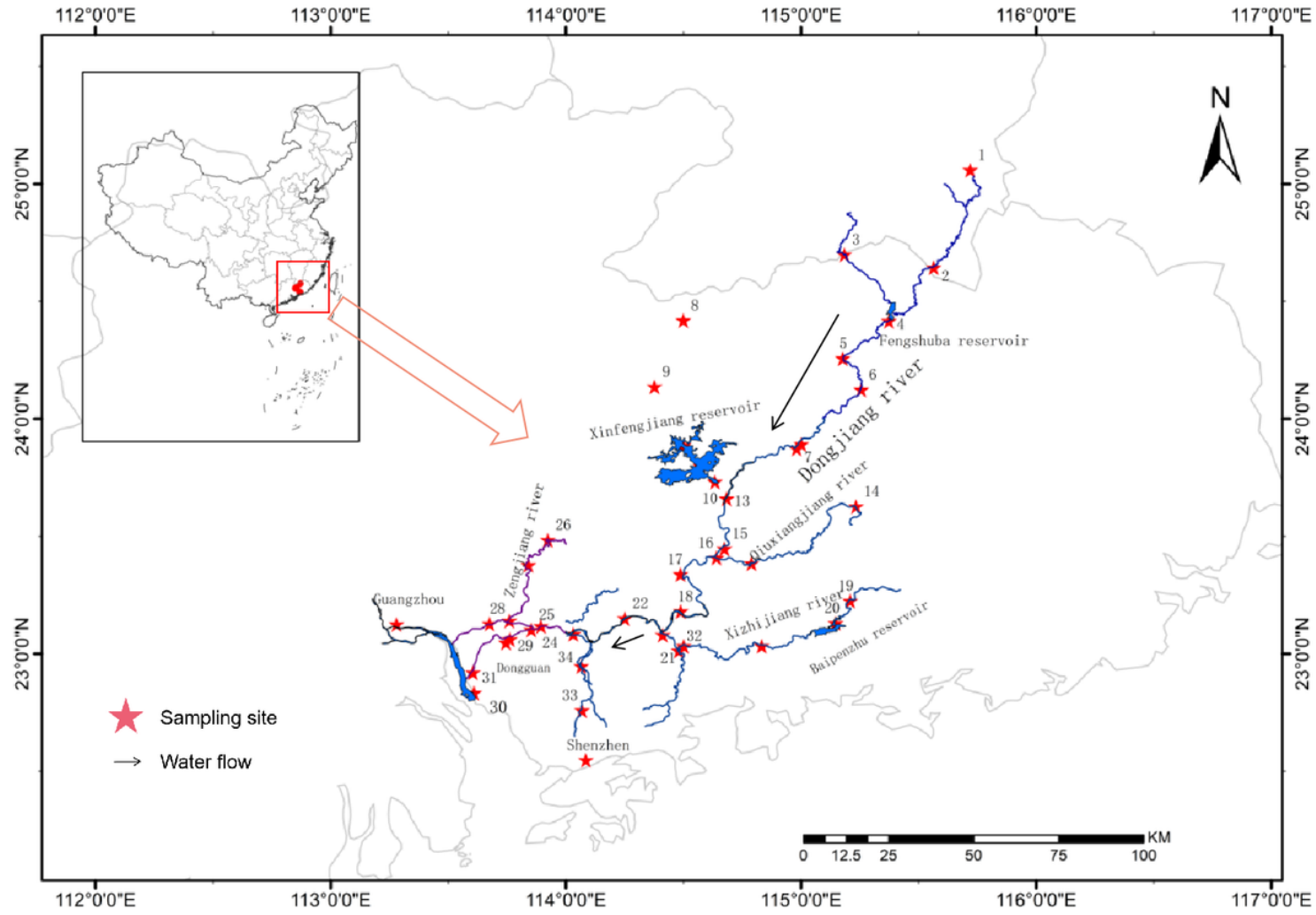


Figure 1

Location map of the sampling sites in the rivers of the Dongjiang River region, South China.

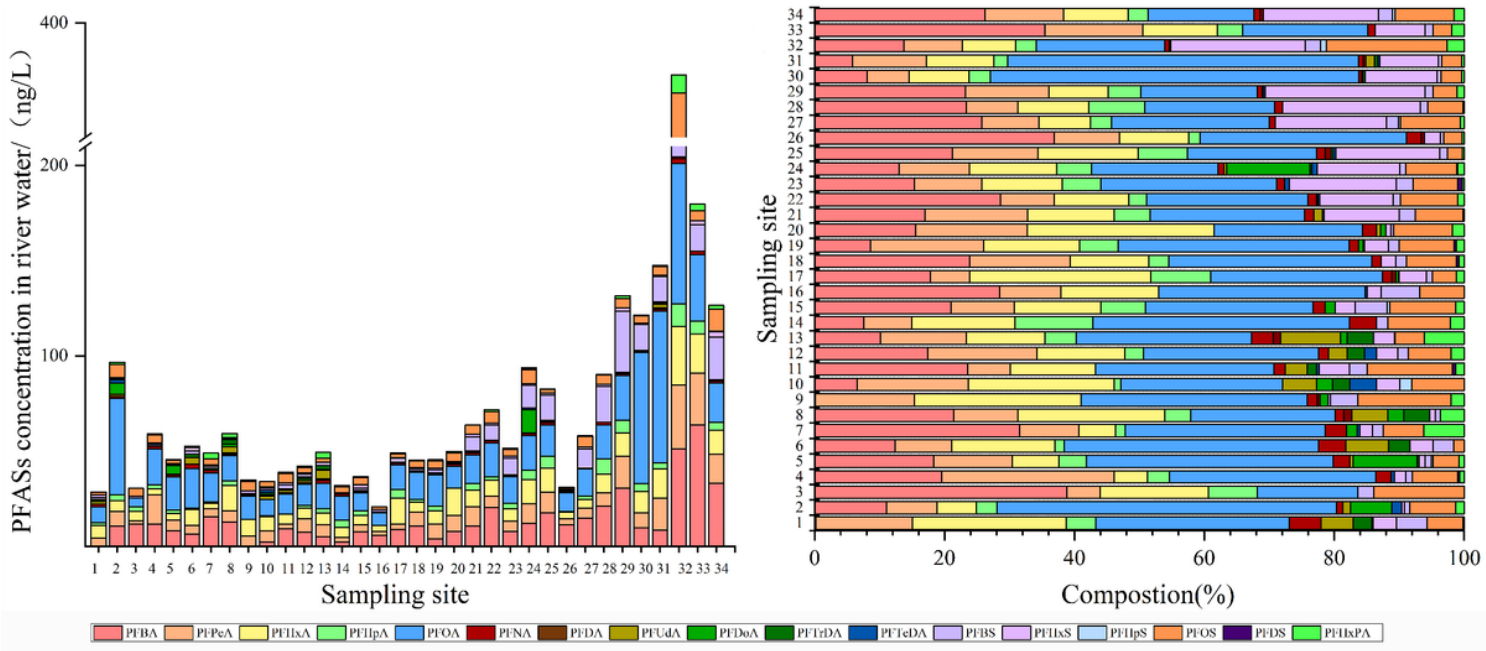


Figure 2

Composition and concentration of target PFASs in water samples in the Dongjiang river.

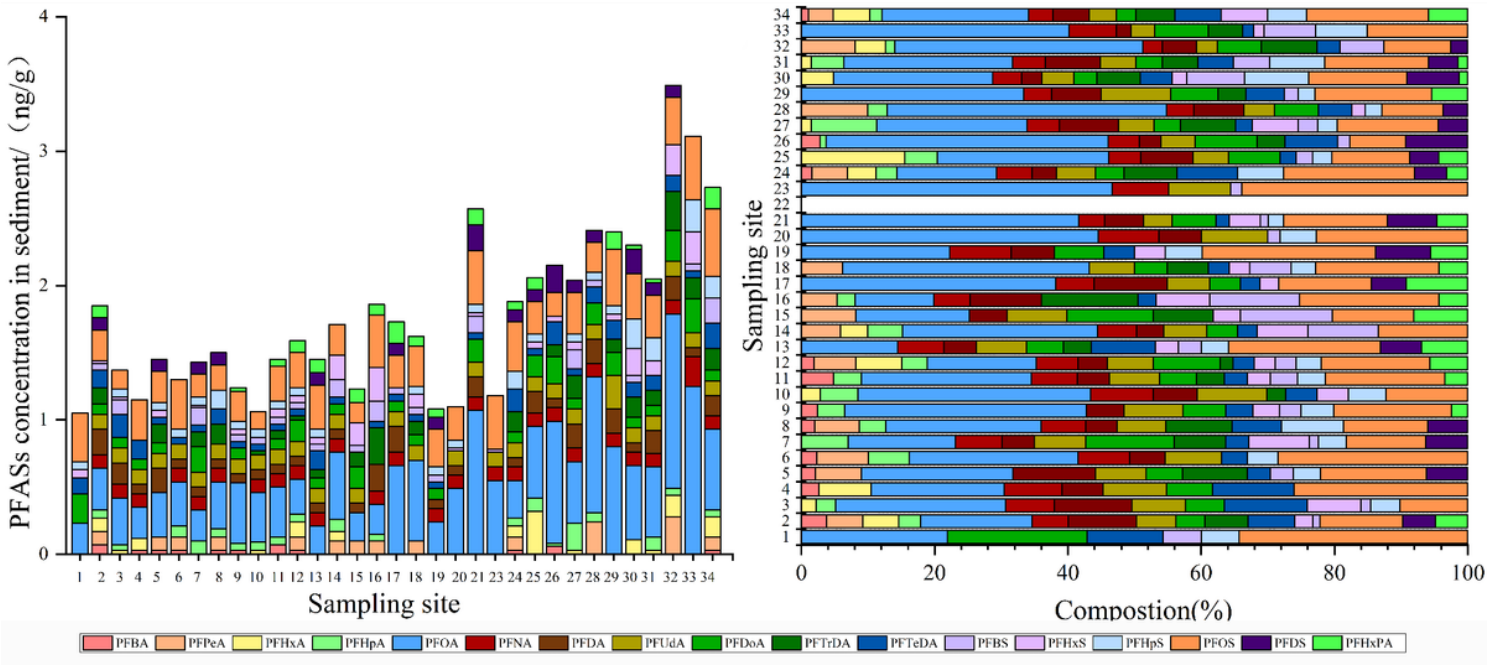


Figure 3

Composition concentration and of target PFASs in sediment samples in the Dongjiang river.

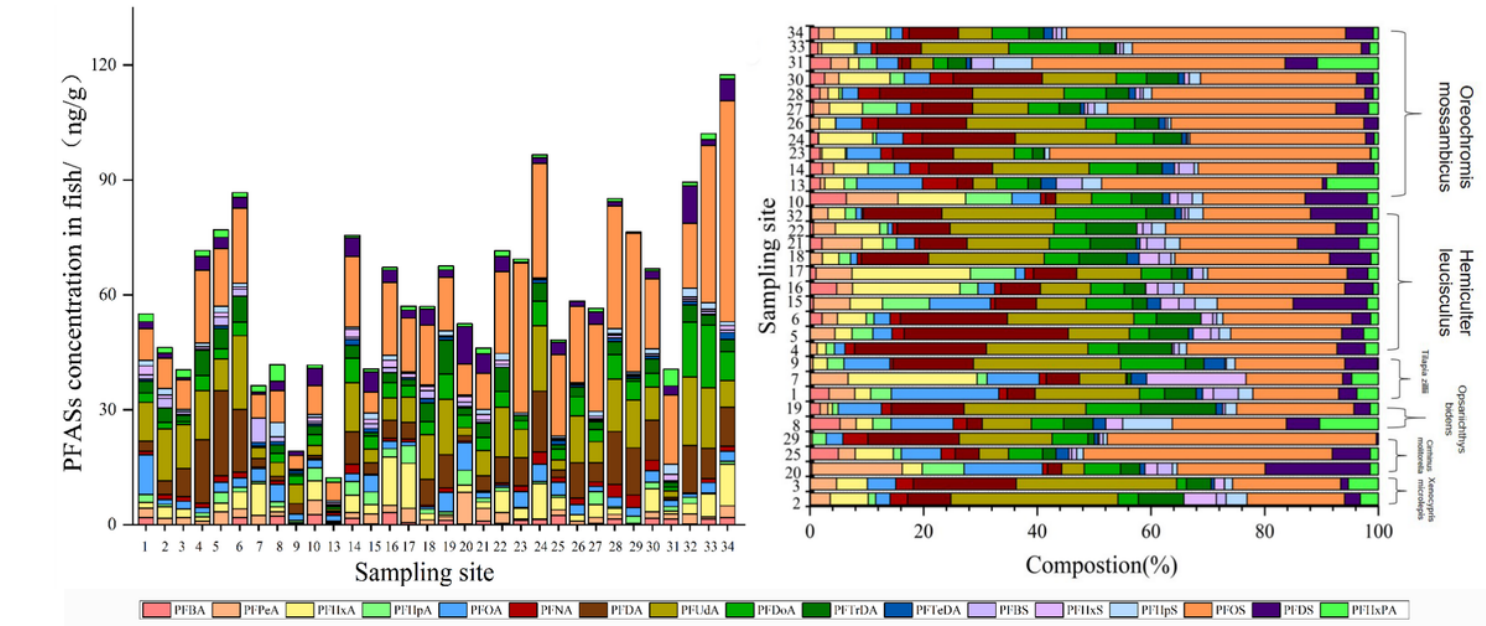


Figure 4

Composition and concentration of target PFASs in fish samples in the Dongjiang river

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